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The Epidemiology of Parkinson's Disease: what we know and what we don't

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Search Strategy

The PubMed and Embase electronic databases were searched up to June 18, 2023, using the following terms and key words: epidemiology, Parkinson's disease, parkinsonism, diagnosis, epidemiology, incidence, prevalence, mortality, distribution, sex, gender, race, racial, socioeconomic, sociodemographic, inequality, disparity, risk factor, protective factor, environment, genetics, gene – environment, toxicant, prevention . Priority was given to reviews published within the past 5 years, supplemented with key articles from the bibliographies of those reviews.

Key words: Parkinson's disease; epidemiology; risk factors, secular trends, prevalence; incidence

Current word length = 4,131; abstract 238

Abstract

The epidemiology of Parkinson's disease (PD) shows marked variations in time, geography, ethnicity, age and sex/gender. Internationally, prevalence has increased over and above demographic changes, and there are several potential reasons why this may be, including the decline in other competing causes of death. There is less certainty about whether incidence is increasing especially for women and for many low middle income countries where there is a shortage of high quality data. PD is more common in older people and men. A variety of environmental factors have been suggested to explain why this may be, including exposure to neurotoxic agents. Within countries there appear to be ethnic differences in disease risk though this may reflect differential access to health care. The aetiology of PD is multifactorial and involves genetic and environmental factors. Both risk (e.g. pesticides) and protective factors (physical activity, liability to smoke) have been postulated though elucidating causality is complicated by the long prodromal latent period. Following the establishment of public health strategies to prevent cardiovascular and some cancers, chronic neurodegenerative diseases such as PD and dementia are gaining a deserved higher priority. Multi-pronged prevention strategies are required that tackle population based primary prevention, high risk targeted secondary prevention as well as PD disease modifying therapies for tertiary prevention. Future international collaborations will be required to triangulate evidence from basic, applied and epidemiological research thereby enhancing the understanding and prevention of PD at a global level.

Introduction

Over 65 years ago, Prof. Jerry Morris, a founding father of epidemiology, stated “One of the most urgent *social* needs of the day is to identify rules of healthy living that might reduce the burden of the metabolic, malignant and ‘degenerative’ diseases....”¹ Epidemiology, complementing biological sciences, provides the scientific basis (i) to monitor the secular patterns of disease to enable future disease projections; (ii) to undertake “community diagnosis” by measuring disease incidence and prevalence to enable health and social care service planning; (iii) to search for risk or protective factors that could be addressed through individual and societal modifications and hence influence prevention strategies. Parkinson’s disease (PD) is the second most common neurodegenerative disease and is a significant societal issue and global priority.^{2,3} In this review of PD epidemiology, we will highlight what we know with confidence, what is less certain and what we still do not know. We cannot discuss all potential risk factors in depth, given word length limitations, so we have chosen those risk factors that we feel are of most interest based on the evidence base, novelty or potential for prevention. Whilst not all readers may agree with our choices, we hope this stimulates further discussion in the academic and policy arenas.

PD Diagnosis

The International Parkinson and Movement Disorders Society (IPMDS) Task Force defined PD as “a core clinical motor syndrome (parkinsonism) accompanied by substantia nigra pars compacta neurodegeneration and synuclein deposition.”⁴ This clinicopathological entity has, to date, been impossible to verify pre-mortem so diagnosis is based on typical parkinsonian motor features in the absence of red flags suggesting alternative diagnoses such as atypical parkinsonian syndromes⁵. This diagnostic approach may change in the forthcoming years, as recent research shows that α -synuclein seed amplification assays can accurately distinguish PD from healthy controls and may also identify the stage of PD.⁶ Because of these findings, novel biomarker-defined definitions of PD are currently being developed to diagnose pre-clinical PD on a biological basis⁷. However, the diagnostic value of biomarker assays to distinguish between PD and atypical parkinsonian syndromes requires further study.

The IPMDS PD Definition Task Force proposed a separate “clinicogenetic” category so individuals carrying highly penetrant mutations can be designated as PD regardless of α -synuclein pathology. Seven genes are currently designated as monogenic causes,⁸ four resulting in later onset, autosomal dominantly inherited disease (*LRRK2*, *CHCHD2*, *VPS35* and *SNCA*) and three in early onset autosomal recessively inherited disease (*PARKIN*, *DJ1* and *PINK1*). Variants in an eighth gene, *GBA*, are the most

common genetic risk factor for PD with penetrance up to 30%⁹. Genetic testing is not a cornerstone of PD diagnosis but helps to clarify prognosis and define risk for family members. The UK Brain Bank Criteria¹⁰ and the more recent updated Movement Disorder Society (MDS) clinical diagnostic criteria¹¹ are the most widely used in clinical care and research to enhance diagnostic accuracy. The sensitivity of the UK Brain Bank criteria is high (over 80%), tested against neuropathological assessment¹⁰. Standard neuropathological benchmarks do not yet exist for PD, and sensitivity may vary if tested against a different standard¹². The 2015 MDS-PD Criteria, based on the UK Brain Bank criteria, have been tested against clinical expert opinion with high accuracy (92.6%), exceeding the UK Brain Bank criteria^{11,13}, but clinicopathological validation studies are awaited. The MDS task force has also defined “prodromal” PD as a true stage of PD recognizing the earlier onset of PD in nondopaminergic neural structures¹⁴. Application of these criteria in prospective population-based studies show high specificity for overt PD development (>98%,)^{15,16}.

There are two types of possible diagnostic error in epidemiological studies. (i) Misdiagnosis (false positives) may result from misclassifying other neurodegenerative pathologies, secondary parkinsonism and non-progressive Parkinson-like conditions, such as dystonic and essential tremor as PD. (ii) Underdiagnosis (false negatives) will also occur if people with PD regard their symptoms as part of “normal aging” and/or do not present, or have no access, to health care as highlighted from door-to-door surveys (see age section below). Even when they do, the diagnosis of PD may be missed when obscured by other co-morbidities such as osteoarthritis, frailty and depression.

Large epidemiological studies frequently use record linkage and electronic health records (e.g. the Rochester Epidemiology Project¹⁷). This can be very accurate if confirmed by specialist medical record review, though administrative codes can over-estimate cases¹⁸. Often studies rely solely on diagnostic codes or drug data without confirmation utilising large databases e.g. insurance claims, or even rely on proxy report of PD¹⁹ at the cost of diagnostic accuracy.

Descriptive epidemiology

Time

With increasing survival into older age, the absolute number of people with PD has and will continue to increase, labelled by some as a “pandemic”²⁰. The Global Burden of Disease (GBD) project estimates that whilst crude prevalence increased by around 74% between 1990 to 2016, the age-standardised prevalence increase, accounting for demographics, was less marked at 22%.²¹ The methods used to estimate global and country specific mortality, prevalence (number of people living

with PD at any time) are complex given the limited high quality country-specific data. In particular the concerns of under-ascertainment of PD mortality meant that the highest mortality to prevalence ratios from three high income countries were used to extrapolate to all other countries, which may not have been generalizable.

There is marked heterogeneity in the observed temporal changes in prevalence and mortality both between and within regions though most showed an increase. Much of the increase is projected to occur in low and middle-income countries, especially in the global south. Suggested reasons are increased longevity; with the decline in mortality risk from cardiovascular²² and other chronic diseases resulting in PD patients more likely to live with their PD (increasing prevalence) and die of PD-related complications as their disease can progress (increased PD-specific mortality). Other possible reasons include increased industrialization, which may explain the largest increases in countries like China, and the reduction in smoking, as rates increased faster for men than women, men being more likely to be smokers and to have quit smoking. This explanation is paradoxical as men also have a higher risk of PD (see below). However, these explanations are unlikely to explain the very large differences reported for countries with similar geographic, demographic and economic characteristics. Why, for example within Europe, does Netherlands have a predicted 7.5% decline whilst Norway have an 87.1% increase in PD prevalence rates?

These analyses did not examine the secular trends in incidence (newly diagnosed PD) as data for this is still very limited. In general, studies using administrative or primary care records find either no increase or a decrease in PD incidence^{23,24}. Two cohorts a decade apart from the population-based Rotterdam study found a similar relative decline in incidence across all age groups²⁵. In contrast, the Rochester Epidemiology Project found a modest increase in PD incidence over a thirty year period that was greater for men than women and for patients over 70 years²⁶. Age-period-cohort models suggested a birth cohort around 1920 for PD in men and predicted that incidence rates should decline in the future. A French cohort of women (E3N) also failed to find any change in incidence over 27 years follow-up²⁷. Even less is known about incidence in less affluent countries though secondary analyses of the GBD project indicate modest increases in the annual percentage change in low (0.23%) and low-middle (0.31%) sociodemographic index countries²⁸. This may reflect better ascertainment and/or a true increase in incidence.

Variations by place, race and ethnicity

Although variations in risk by geography (“place”), race and ethnicity are typically discussed separately, in many parts of the world these characteristics may not be easily differentiated. In addition, racial/ethnic origin are socio-politically defined and heavily influenced by differences in access to education, health care, socio-economic status and other factors²⁹. Inequities across these groups complicates efforts to better explore biological and environment effects on disparities in health status.

Incidence estimates are only available for limited, usually HICs, regions. Prevalence estimates vary considerably across geographical regions.^{21,30-36} Compared to Europe and North America, the prevalence of PD has been estimated to be lower in Africa (in particular in Sub-Saharan Africa)^{37,38}, similar^{39,40} or lower in Asia^{2,21}, and similar in Latin America^{36,41-46}. There are a variety of potential, but unproven, explanations for the observed heterogeneity in prevalence including geographical differences in (i) genetic variations⁴⁷ (ii) environmental factors, (iii) gene-environment interactions (iv) access to health care, (v) life expectancy, (vi) study methodology. Occupational and passive exposure to pesticides may contribute to the rise in low-middle-income-countries (LMICs), whilst air pollution may be more important in industrialized high-income-countries (HICs)

Some clues exist to the inter-relationship between genetic and environmental factors and PD risk. (i) International, multi-ethnic genetic studies show that the frequency^{48,49} and penetrance⁵⁰ of genetic risk variants of PD varies across ethnic groups and geographical regions, suggesting that genetic factors account for at least some of the heterogeneity. (ii) Migrant studies show that estimates of PD prevalence are considerably higher in people of African origin living in the United States than in Black-Africans residing in sub-Saharan Africa.⁵¹ Similarly, the prevalence and incidence of PD for Japanese-American men living in Hawaii are higher than for Japanese men living in Japan.⁵² Such observations might be explained by differential exposure to a wide range of adverse exposures (e.g. occupational, poverty-related etc) in migrant populations. Alternatively, the non-migrant population might have less awareness of the disease or access to health care leading to underdiagnosis. (iii) Despite generally lower rates of Parkinsonism in Blacks compared to Whites, two post-mortem studies found similar estimates of Lewy body pathology in White and Black individuals.^{53,54} This suggests that sub-clinical disease risk may be similar but either slower progression or worse access to health care results in less clinical diagnosis in Blacks. (iv) Studies on ethnically diverse individuals living in the same area show that the incidence and prevalence of PD to be higher in European White people than in Black people or Asian people.^{30,55-58} A US study reported the incidence of PD to be higher among people of Hispanic ethnicity as compared to other non-Hispanic whites than among

Black, Asian⁵⁵, although another study in New York City noted higher incidence just in Black men but lower prevalence of PD in Black than White people, though they did not differentiate White groups due to small numbers³⁸. It is unclear whether these differences are real or due to bias in case-ascertainment through differential access to health care.

Most of the above mentioned studies have been carried out in settings where health and social disparities among different race/ethnic groups potentially bias the results. In addition, self-declared race/ethnicity has been used as a surrogate for genetic ancestry despite being more reflective of social-cultural conditions. Future studies will need to distinguish genetic/biological effects from environmental/social effects. Such an approach could provide valuable clues to modifiable environmental factors. For example, comparing risk of PD in a large country like as India which varies both geographically (North, South) as well as by race/ethnicity usually proxied by religion (e.g. Hindu, Muslim, Jain, Sikh) could be valuable and has been studied with other major disease groups⁵⁹.

Age

There is no controversy that increasing age is associated with increased PD risk though whether this is a linear or exponential increase remains unclear. Unless individuals are screened, older individuals will be under-diagnosed to a greater degree (see above). In the EUROPARKINSON study, the percentage of undiagnosed screen-detected cases increased from 18% for 65 to 70 years to 36% for those aged 80 to 85 years³¹. This further increases to over 50% of people over 85 years in a sample of older adults living in East Boston, Massachusetts, if one uses the criteria of two or more features of parkinsonism (bradykinesia, gait disturbance, rigidity, tremor) rather than clinically diagnosed PD⁶⁰. This age pattern is similar to that seen for other neurodegenerative disorders and many cancers. Whether age-related cell death is an inevitable function of biological ageing or is secondary to chronic exposure to environmental toxins is unknown. In most cases, complex interactions among senescence-associated cellular changes, genetically-determined processes and environmental insults likely underlie the neurodegenerative process⁶¹.

Sex

The incidence, prevalence and mortality risk of Parkinson's disease is higher in men than women by a ratio of approximately 1.4:1⁶², and the prevalence ratio estimate has remained stable for the past two to three decades²¹. The incidence ratio, may have increased over time; The Rochester project found incidence in men increasing over the interval between 1976-1985 and 1996-2005 but remaining stable in women²⁶. The reason for higher incidence in men is not fully understood but the

observation that in LRRK2-associated PD, there is, if anything a higher risk in women with the G2019S mutation supports the role of environmental factors⁶³. Hypotheses include men being more exposed to adverse environmental risk factors or the protective role of female hormones. However unlike cardiovascular disease, where the male to female risk ratio disappears after 75 years, women appear to have lower risk at all ages. Similar to race, few studies try to distinguish how sex (biological) may differ from gender identity (sociocultural) in terms of PD risk.

Socioeconomic status (SES)

There are marked disparities between socioeconomic status (SES) and disease risk with poorer individuals, in general, having a higher risk reflecting greater exposure to a wide range of adverse exposures across the life course. Studies that have assessed the association between SES and PD find inconsistent results. This is surprising as higher SES in HICs is associated with a lower chance of being a current or ex-smoker (see below). Recent Mendelian randomization analyses find that genetic variants (SNPs) that predict higher cognition were associated with increased PD risk⁶⁴, the opposite to what is observed for Alzheimer's dementia⁶⁵. Possibly the failure to find a consistent SES gradient reflects that people of lower SES are often more likely to be exposed to neurotoxic exposures countering the expected positive gradient.

Determinants of disease

In trying to identify causal determinants of disease, it is important to triangulate knowledge obtained in basic and clinical science as well as epidemiologic studies⁶⁶. Disease determinants can be intrinsic or extrinsic (see figure 1). Possible extrinsic factors include environmental agents (toxicants, infectious organisms), lifestyle behaviours such as diet and physical factors such as trauma. Intrinsic factors include genetic predisposition, metabolic state (e.g., elevated urate) or co-morbidities (e.g., diabetes mellitus). Intrinsic factors may result from or interact with extrinsic factors, including epigenetic changes, the microbiome or other comorbidities. Some determinants may affect disease risk in men and women differently (see Table 1).

It is generally accepted that the majority of PD cases result from the combined effects of environmental exposures and variation in genes regulating metabolic pathways, such as detoxification genes which contribute to susceptibility for PD⁶⁷. Pathogenic variants in a number of genes are associated with familial PD, representing the minority of PD cases (see above). For most of these variants, penetrance is incomplete, and other genetic and/or environmental factors likely determine if and when disease will manifest.

Concordance rates in monozygotic and dizygotic twins with lifelong follow-up estimated the heritability of PD to be only 30%, supporting a larger role for environmental and behavioural factors⁶⁸. Because both genetic and environmental factors must be measured in order to assess for gene-environment interaction, few studies have yet been performed, usually in small samples. One exception is from UK BIOBANK which examined for interactions between a PD polygenic risk score and a very limited range of risk factors¹⁹. They found that only the adverse effect of type II diabetes was stronger in those with less genetic predisposition. This illustrates how the contribution of extrinsic factors will differ conditional on genetic risk. Nearly all studies classify PD cases together or possibly stratify by known genetic variants, ignoring heterogeneity (“clusters”) in PD phenotype. Only one study has examined whether PD genetic risk score predicts variations in phenotype. This found that a more severe phenotype, previously shown to be associated with a pro-inflammatory profile, had a lower genetic risk than other clusters and may be more driven by extrinsic factors⁶⁹, but requires replication. This has important methodological implications if such participants are less likely to be included in studies (“selection bias”).

A recent umbrella review of meta-analyses and systematic reviews concluded that constipation, a well-recognized prodromal feature of PD associated with Lewy pathology, and physical activity were the only factors providing Class I evidence of association with PD risk⁷⁰. Systematic reviews such as this one are useful in identifying associations easily measured in large populations but will omit more novel risk factors with limited published literature. Further, many environmental factors proposed to be associated with PD are still difficult to measure at a large scale and thus will be under-represented in such reviews. We briefly summarise some risk factors that in our opinions may guide disease prevention strategies, if genuinely causal.

The most consistent association, recognised over five decades ago, is a reduced risk of PD in cigarette smokers and, in a few studies, other tobacco users, though its interpretation has remained controversial⁷¹. The association shows a dose-response effect being stronger with longer duration and greater frequency of tobacco use. Mendelian randomization studies, which avoid reverse causation and reduce confounding, support this association. Genetic variants that determine smoking liability are associated with lower risk of PD⁷². The mechanism of this association is not known. Nicotine may or may not play a central role, though trial evidence of disease modifying effects of the nicotine patches in PD patients is absent⁶⁷. Coffee and tea drinking are also associated with a lower risk of PD, particularly in men^{73,74}. Dietary dairy intake is associated with greater risk of

PD, possibly due to toxicants bioconcentrated in milk⁷⁵, while diets higher in vegetables, fruits and grains are associated with a reduced risk of PD though this may reflect residual confounding by other factors⁷⁶. Anti-inflammatory drug use is associated with a decrease in risk in PD⁷⁷, including that associated with LRRK2⁷⁸. Inflammation is thought to contribute to PD pathogenesis in sporadic and familial disease. Physical activity and exercise are associated with lower risk of PD, and greater amounts and intensity of exercise are associated with greater risk reduction, although even modest levels reduce risk⁷⁹. The combined effects of these lifestyle factors appear to be additive, suggesting an approach to disease prevention⁷⁶ analogous to cardiovascular disease.

Environmental toxicant exposures, including pesticides, solvents and air pollution, are associated with higher risk of PD. PD risk is increased in persons with occupational pesticide use or those living near pesticide treated fields, reducing recall bias⁸⁰. PD-associated pesticides including paraquat, rotenone, 2,4-D and several organochlorines have biochemical effects including mitochondrial dysfunction, inflammation, epigenetic methylation and alterations of the microbiome that are thought to be important to PD (see Morris companion article), strengthening the possibility that these associations are causal effects^{61,80,81}. Genetically determined impairment in toxicant handling can increase the risk of PD due to pesticide exposure, while extrinsic factors such as use of personal protective equipment when using pesticides and eating a healthy diet can lower risk in those exposed. Chlorinated solvents (trichloroethylene, perchloroethylene, carbon tetrachloride) are associated with increased PD risk in humans and cause parkinsonism-associated toxicity in animal models⁸². Many PD-associated toxicant chemicals are environmentally persistent, which is relevant given the long latency period between exposure and disease diagnosis (see below on life course exposures). For example, trichloroethylene can be detected in nearly one third of US drinking water supplies, in soil, air, food and human breast milk. Other toxicant exposures associated with PD with include air pollution and occupational exposure to welding or certain metals⁸³.

Meta-analysis of case control studies show an association of head injury with PD⁸⁴ but a record linkage study of hospital admissions, avoiding recall bias, and PD risk only found an association close to diagnosis suggesting reverse causation⁸⁵. In contrast, a high quality retrospective cohort study with prior documented traumatic brain injury in the Veterans Health Administration database found a dose-response relationship with severity⁸⁶. The sequelae of head trauma may also depend on genetic susceptibility, such as variants in the gene encoding alpha synuclein⁸⁷. Infectious agents, and particularly viral agents, may also serve as risk factors for PD, though the evidence for this is not

consistent. Such exposure, in susceptible individuals, may initiate an inflammatory process increasing vulnerability to other risk factors or triggering a neuropathological cascade⁸⁸.

Prevention of PD

Prevention strategies can be classified as primary, secondary and tertiary prevention (see figure 2). Primary approaches are population based and intervene for the whole population regardless of risk and may operate before degeneration begins (2a). They often require political or structural changes, such as banning a neurotoxic agent. Secondary interventions target people at greater risk of developing PD using either genetic risk markers or prodromal symptoms such as impaired olfaction, REM sleep behaviour disorder and constipation⁸⁹. Intervention in the prodromal phase (2b) may either delay cellular degeneration or slow its rate of progression. Any pharmacological intervention will need to balance the cost and risk of any intervention given the relatively high number of people who might be treated that will never develop clinical PD. Tertiary prevention treat people with established disease with disease modifying therapies that slow or halt progression and hence reduces future PD complications and PD mortality (2c).

It has become increasingly recognised that many chronic diseases and ageing phenotypes have both developmental as well as degenerative phases and should be viewed within a “life course” framework⁹⁰ (figure 2 is a hypothetical schematic). The possible developmental origins of PD have been relatively understudied⁹¹. There is an absence of neuropathological evidence at the moment. Possible aetiological factors that may act in the healthy phase (2a) include childhood infections, neurotoxic exposure, smoking behaviour and developmental factors such as height and weight gain. Understanding the timing of any toxic exposures may be particularly important in relation to any critical or sensitive periods whereby the brain may be more or less susceptible to any adverse impact. Delineating the prodromal or latency period is also important to determine if exposures either reflect reverse causation secondary to the existing pathophysiology or are accelerate the manifestation of clinical disease after onset of pathology that rather than initiating it.

The need for secondary prevention trials

Interventions in the prodromal phase of PD may be more effective than agents targeting disease mechanisms after disease has become established, which to now have been disappointing, similar to that seen with therapeutic trials for Alzheimer’s disease⁹². The Michael J Fox Foundation has recently announced plans to assess therapeutics in persons who have clinical, imaging and biomarker

characteristics of prodromal Parkinson's disease who are participating in the longitudinal PPMI study⁹³.

One promising non-pharmacological intervention is promoting physical activity in people with prodromal features of PD, such as idiopathic REM sleep behaviour disorder (RBD). There is converging evidence that intense physical activity may exert disease-modifying properties in PD, whilst the risk of side effects and costs are low⁹⁴. A serious challenge, however, will be to sustain compliance with any exercise intervention given the highly variable latency period between the onset of RBD and PD which may span more than a decade⁹⁵. New digital technologies enable a cheap cost-effective way to deliver a self-directed intervention, using gamification, as well as remote measurement of intermediate endpoints as well as PD phenoconversion.

Applying such an approach at a population level could lead to large benefits, preventing 14.5% of all current cases given the following assumptions; (i) the lifetime risk of PD is approximately 5%, (ii) the summary risk ratio for PD comparing those with moderate to vigorous physical activity versus low levels is 0.71⁹⁶ and this reflects a causal effect (iii) 50% of the general population do not meet the World Health Organisation recommended standard for moderate to vigorous physical activity⁹⁷, (iv) societal interventions could abolish low activity levels. Whether this is or is not achievable in the real world, it demonstrates the large prevention potential of a population-based approach. Albeit an extreme example, the US embargo of Cuba between 1990 to 1995 led to an increase of physical activity from around 30% to 80% as well as decline of calorie intake between 3,200 to 2,400 Kcal with a corresponding weight loss of 5.5 Kg and 33% decline of diabetes incidence (1.8 per 1000 to 1.2 per 1000)⁹⁸.

Conclusions

We have attempted to summarise what we know and don't know about the epidemiology of PD, how to prevent it and future research needs (see table 1). A recent paper by a World Health Organization task force similarly highlighted the need for better quality epidemiological data with equitable representation (by race, ethnicity, geography, sex, and gender) and the lack of harmonised risk reduction and prevention strategies⁹⁹. In LMICs, there was also a need to enhance research infrastructure, better training and support, appropriate governance procedures and greater inclusion of civil society organizations and people with PD. Even in HICs there often remain too many silos between basic, clinical and epidemiological researchers who often view each as competitors for limited funding. We welcome initiatives such as "Aligning Science Across Parkinson's (ASAP)" which

aims to foster collaboration and promotes data sharing to promote new discoveries with programs such as GP2 though its focus is around genetics. We hope future endeavours will aim to measure global environmental factors and use natural experiments to promote evidence based prevention strategies.

Contributors

YBS and CT had the original conceptualization which was then discussed with all co-authors and agreed. Each author lead on various sub-sections of the paper that were then collated and harmonised by YBS. Each iteration of the paper was reviewed and discussed by all co-authors who all helped in drafting the response to the reviewers, revise and approve the final manuscript.

Declaration of Interests

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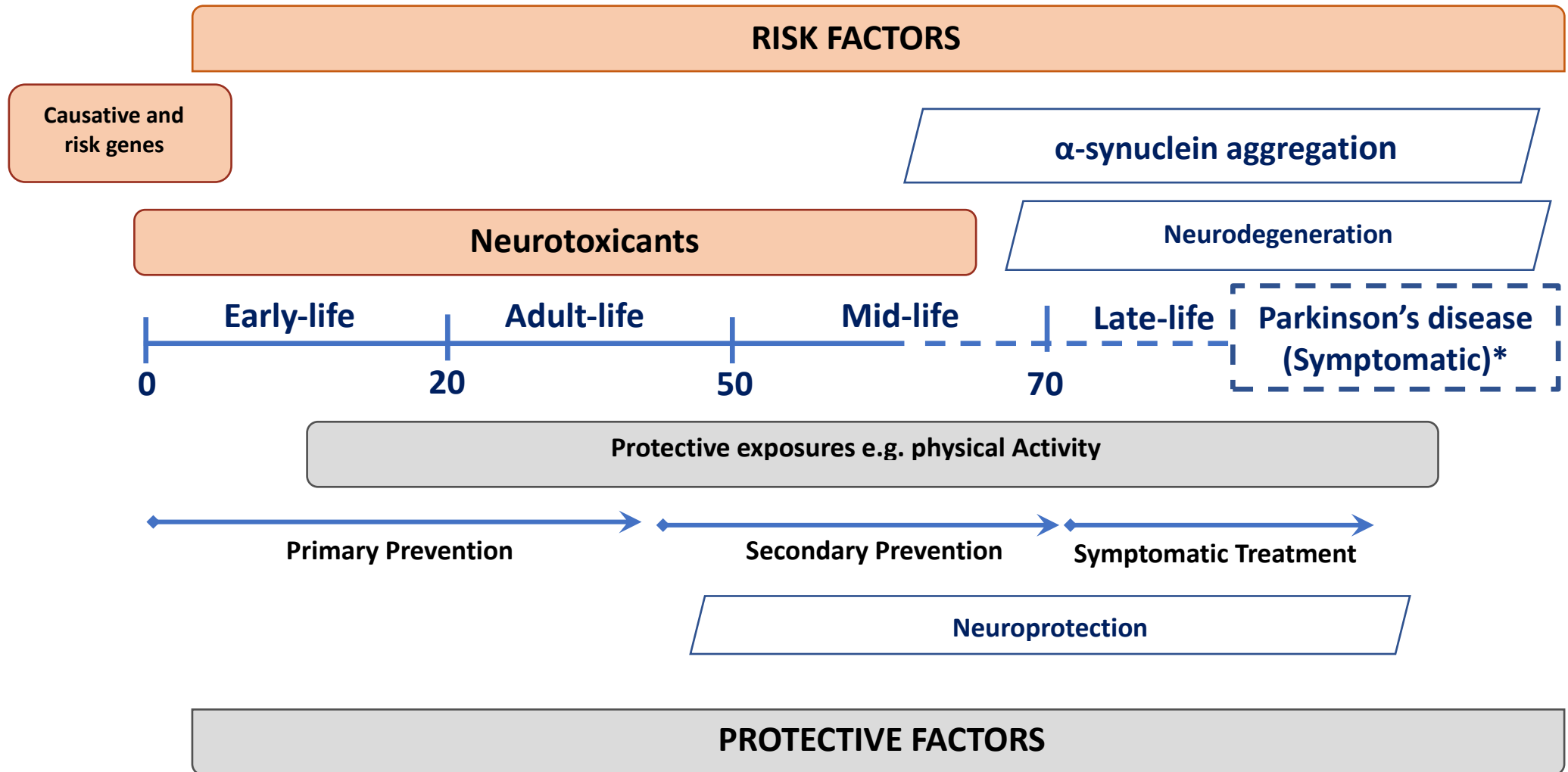
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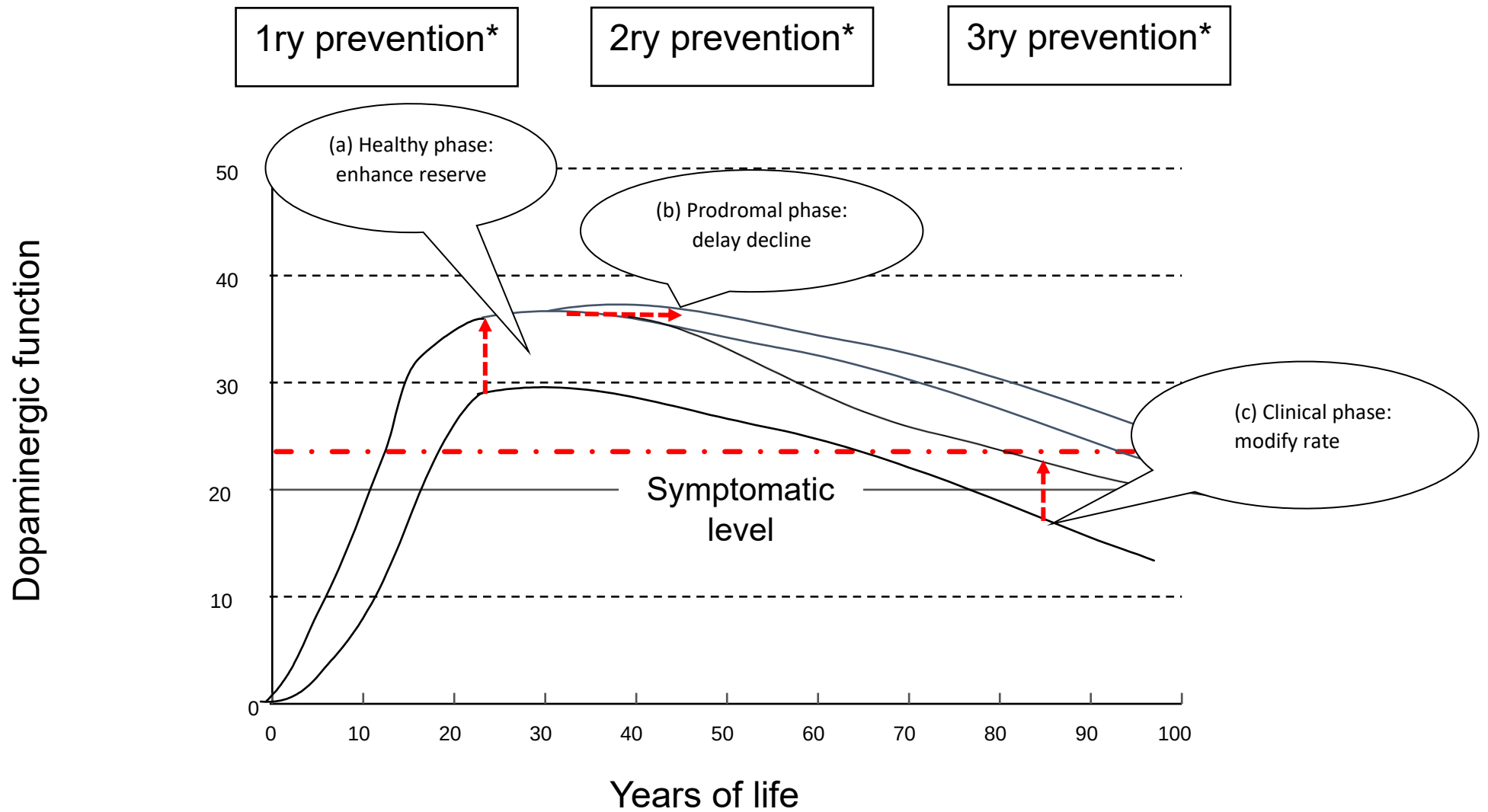
Figure 1: Simplified schematic of how risk and protective factors may determine neuropathology and the development of PD

* Whilst PD is shown as developing in later life, the age of onset is variable and there are cases that develop disease at much younger ages.

Figure 2: Hypothetical primary, secondary and tertiary preventative strategies to prevent, slow the onset and slow the progression of PD.

* No assumptions have been made about the relative effectiveness of each strategy





Prevention strategies across the life course

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